Graph Embeddings

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Introduction to Graph Embeddings

- Focus: Data with semantic relationships between training samples $\{\mathbf{x}_n\}_{n=1}^{N}$.
- Relationships (edges) connect training samples (nodes) with application-specific meaning (commonly similarity).
- Graphs provide mathematical foundations for reasoning about these relationships.

Graphs as Universal Data Structures

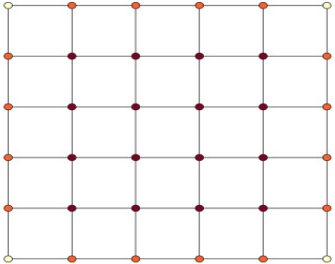
- Represent complex relational data (nodes and edges).
- Appear in multiple domains:
  - Social networks
  - Computational chemistry [Gil+17]
  - Biology [Sta+06]
  - Recommendation systems [KSJ09]
  - Semi-supervised learning [GB18]
- Notation:
  - $\mathbf{A} \in \{0,1\}^{N \times N}$: Adjacency matrix ($N$ = number of nodes).
  - $\mathbf{W} \in \mathbb{R}^{N \times N}$: Weighted version of adjacency matrix.
  - $\mathbf{X} \in \mathbb{R}^{N \times D}$: Matrix of node features.

Challenges in Graph Data

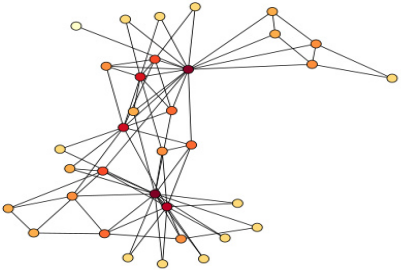
- Neural networks on graphs should apply to nodes in different graph settings (varying connections, community structure).
- Contrast with images: each pixel (node) has the same neighborhood structure.
- Arbitrary graphs: no specified alignment of nodes, each node might have different neighborhood structure.
- Euclidean spatial convolution cannot be directly applied on irregular graphs.
  - Euclidean convolutions rely on geometric priors (e.g., shift invariance) not generalizable to non-Euclidean domains.

Geometric Deep Learning (GDL)

- Developed to address challenges of applying deep learning to non-Euclidean data [Bro+17b].
- Graph Representation Learning (GRL) [Cha+21]:
  - Aims to learn low-dimensional continuous vector representations (embeddings) for graph-structured data.
- Figure 23.1: Euclidean vs. non-Euclidean graphs.



(a) Grid (Euclidean).



(b) Arbitrary graph (Non-Euclidean)

Classes of GRL Problems

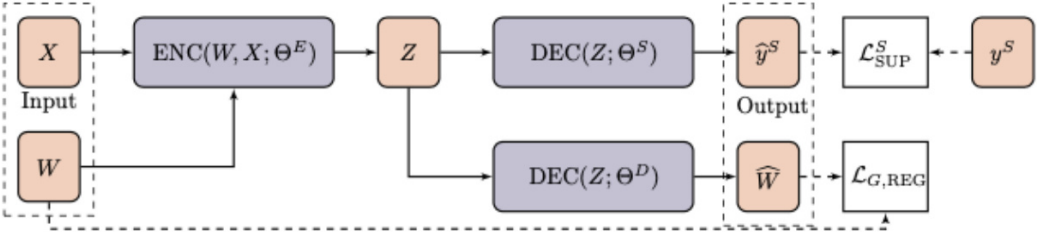
- **Unsupervised GRL**:
  - Learns low-dimensional Euclidean representations.
  - Objective: Preserve input graph structure.
- **Supervised (or Semi-supervised) GRL**:
  - Learns low-dimensional Euclidean representations.
  - Objective: For a specific downstream prediction task (node or graph classification).
- **Learning Settings**:
  - *Transductive*: Graph structure is fixed during training and testing (e.g., predicting user properties in a social network).
  - *Inductive*: Model answers questions about graphs not seen during training (e.g., classifying molecular structures).
- **Embedding Spaces**:
  - Most methods learn Euclidean vector spaces.
  - Recent interest in non-Euclidean representation learning (e.g., hyperbolic, spherical spaces).
  - Motivation: Use continuous embedding space resembling the discrete structure of input data (e.g., hyperbolic space for trees [Sar11]).

Graph Embedding as an Encoder/Decoder Problem

- Many GRL methods follow a similar pattern:
  1. Encode input (node features $\mathbf{X} \in \mathbb{R}^{N \times D}$ and graph edges $\mathbf{A}$ or $\mathbf{W} \in \mathbb{R}^{N \times N}$) from discrete domain to continuous representation (embedding) $\mathbf{Z} \in \mathbb{R}^{N \times L}$.
  2. Use learned representation $\mathbf{Z}$ to optimize an objective (e.g., reconstructing graph links).
- We use the Graph Encoder-Decoder Model (GraphEDM) by Chami et al. [Cha+21] to analyze GRL methods.

GraphEDM Framework

- Figure 23.2: Illustration of the GraphEDM framework [Cha+21].



- Encapsulates supervised and unsupervised methods, including:
  - Graph as regularizer [ZG02]
  - Positional embeddings [PARS14]
  - Graph neural networks (message passing [Gil+17; Sca+09], graph convolutions [Bru+14; KW16a])

GraphEDM Components: Input

- Input:
  - Weighted graph $\mathbf{W} \in \mathbb{R}^{N \times N}$
  - Optional node features $\mathbf{X} \in \mathbb{R}^{N \times D}$
- (Semi-)supervised settings: Training target labels for nodes (N), edges (E), or entire graph (G).
- Supervision signal $S \in \{N, E, G\}$.

GraphEDM Components: Networks

- **Graph encoder network** $\mathrm{ENC}_{\Theta^E} : \mathbb{R}^{N\times N} \times \mathbb{R}^{N\times D} \to \mathbb{R}^{N\times L}$:
  - Parameterized by Θ^E.
  - Combines graph structure and optional node features.
  - Produces node embedding matrix $\mathbf{Z} \in \mathbb{R}^{N\times L}$:

$$\mathbf{Z} = \mathrm{ENC}(\mathbf{W}, \mathbf{X}; \Theta^E)$$

  - Captures different graph properties depending on supervision.

GraphEDM Components: Networks (Cont.)

- **Graph decoder network** $\mathrm{DEC}_{\Theta^D} : \mathbb{R}^{N \times L} \to \mathbb{R}^{N \times N}$:
  - Parameterized by Θ^D.
  - Uses node embeddings $\mathbf{Z}$ to compute similarity scores for all node pairs in matrix $\widehat{\mathbf{W}} \in \mathbb{R}^{N \times N}$:

$$\widehat{\mathbf{W}} = \mathrm{DEC}(\mathbf{Z}; \boldsymbol{\Theta}^D)$$

- **Classification network** $\mathrm{DEC}_{\Theta^S} : \mathbb{R}^{N \times L} \to \mathbb{R}^{N \times |\mathcal{V}|}$ (where $\mathcal{V}$ is label space):
  - Used in (semi-)supervised settings, parameterized by Θ^S.
  - Output: distribution over labels $\hat{y}^S$, using node embeddings:

$$\widehat{\boldsymbol{y}}^S = \mathrm{DEC}(\mathbf{Z}; \boldsymbol{\Theta}^S)$$

- Specific choices of encoder/decoder networks realize specific graph embedding methods.

GraphEDM: Output and Loss Function

- Output:
  - Reconstructed graph similarity matrix $\widehat{W}$ (unsupervised).
  - Labels $\widehat{y}^S$ for supervised applications (node, edge, or graph level).
- Loss function optimizes parameters $\Theta = \{\Theta^E, \Theta^D, \Theta^S\}$:

$$\mathcal{L} = \alpha \mathcal{L}^S_{\text{SUP}}(y^S, \hat{y}^S; \mathbf{\Theta}) + \beta \mathcal{L}_{\text{G,RECON}}(\mathbf{W}, \widehat{\mathbf{W}}; \mathbf{\Theta}) + \gamma \mathcal{L}_{\text{REG}}(\mathbf{\Theta})$$

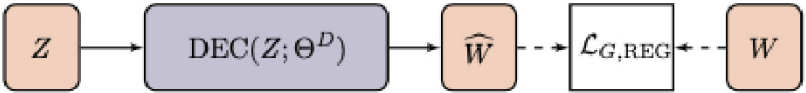
- α, β, γ are hyperparameters.
- Training can be supervised ($\alpha \neq 0$) or unsupervised ($\alpha = 0$).
  - Supervised: Leverages additional information (node/graph labels).
  - Unsupervised: Relies only on graph structure.

Shallow Graph Embeddings

- Transductive graph embedding methods.
- Encoder function maps categorical node IDs to a Euclidean space via an embedding matrix.
- Each node $v_i \in V$ has a low-dimensional learnable embedding vector $\mathbf{Z}_i \in \mathbb{R}^L$.
- Shallow encoder function:

$$\mathbf{Z} = \mathrm{ENC}(\Theta^E) \triangleq \Theta^E \quad \text{where} \quad \Theta^E \in \mathbb{R}^{N \times L}$$

- Embedding dictionary $\mathbf{Z}$ is directly learned as model parameters.
- **Unsupervised case**: Embeddings $\mathbf{Z}$ optimized to recover information about the input graph (e.g., adjacency matrix $\mathbf{W}$ or its transformation). Similar to dimensionality reduction (e.g., PCA) but for graph data.
- **Supervised case**: Embeddings optimized to predict labels (nodes, edges, or whole graph).
- Figure 23.3: Shallow embedding methods. Encoder is an embedding look-up; graph structure used only in the loss.



Unsupervised Shallow Embeddings: Types

- Two main types:

1. **Distance-based methods**:
   - Optimize embedding dictionary $\mathbf{Z} = \Theta^E \in \mathbb{R}^{N \times L}$.
   - Goal: Nodes i, j close in the graph (by graph distance) are embedded such that $d_2(\mathbf{Z}_i, \mathbf{Z}_j)$ is small.
   - $d_2(\cdot, \cdot)$ is a pairwise distance function between embedding vectors (can be Euclidean or non-Euclidean).
   - Decoder output: $\widehat{\mathbf{W}} = \mathrm{DEC}(\mathbf{Z}; \mathbf{\Theta}^D)$, with $\widehat{W}_{ij} = d_2(\mathbf{Z}_i, \mathbf{Z}_j)$.
2. **Outer product-based methods**:
   - Rely on pairwise dot products for node similarities.
   - Decoder network: $\widehat{\mathbf{W}} = \mathrm{DEC}(\mathbf{Z}; \Theta^D) = \mathbf{Z}\mathbf{Z}^\top$.

Reconstruction loss function:

$$\mathcal{L}_{\mathrm{G,RECON}}(\mathbf{W}, \widehat{\mathbf{W}}; \mathbf{\Theta}) = d_1(s(\mathbf{W}), \widehat{\mathbf{W}})$$

- $s(\mathbf{W})$: optional transformation of adjacency matrix $\mathbf{W}$.
- d_1: pairwise distance function between matrices (not necessarily same as d_2).
- Examples: $s(\mathbf{W}) = \mathbf{W}$ or $s(\mathbf{W}) = \mathbf{W}^2$. If $\mathbf{W} = \mathbf{A}$ (binary), use $s(\mathbf{W}) = 1 - \mathbf{W}$ to set connected nodes' distance to 0.

Shallow Embeddings: Distance-based Euclidean Methods

- Minimize Euclidean distances between similar (connected) nodes.
- **Multi-dimensional Scaling (MDS, Sec. 20.4.4)**:
  - $s(\mathbf{W})$: distance matrix (e.g., shortest path distances).
  - $d_1(s(\mathbf{W}), \widehat{\mathbf{W}}) = \sum_{i,j}(s(\mathbf{W})_{ij} - \widehat{\mathbf{W}}_{ij})^2 = \|s(\mathbf{W}) - \widehat{\mathbf{W}}\|_F^2$
  - $\widehat{\mathbf{W}}_{ij} = d_2(\mathbf{Z}_i, \mathbf{Z}_j) = \|\mathbf{Z}_i - \mathbf{Z}_j\|$
- **Laplacian Eigenmaps (Sec. 20.4.9)**: Solves generalized eigenvector problem:

$$\min_{\mathbf{Z}\in\mathbb{R}^{|V|\times d}} \operatorname{tr}(\mathbf{Z}^\top \mathbf{L}\boldsymbol{\Lambda}\mathbf{Z}) \quad \text{s.t.} \quad \mathbf{Z}^\top\mathbf{D}\mathbf{Z} = \mathbf{I}, \quad \mathbf{Z}^\top\mathbf{D}\mathbf{1} = \mathbf{0}$$

- where $\mathbf{L} = \mathbf{D} - \mathbf{W}$ is graph Laplacian, and $\mathbf{D}$ is the degree matrix.
- Equivalent: $\operatorname{tr}(\mathbf{Z}^\top\mathbf{L}\mathbf{Z}) = \frac{1}{2}\sum_{i,j} W_{ij}\|\mathbf{Z}_i - \mathbf{Z}_j\|_2^2$
- Reconstruction: $d_1(s(\mathbf{W}), \widehat{\mathbf{W}}) = \sum_{i,j}\mathbf{W}_{ij}\cdot\widehat{\mathbf{W}}_{ij}$, with $\widehat{\mathbf{W}}_{ij} = \|\mathbf{Z}_i - \mathbf{Z}_j\|_2^2$, $s(\mathbf{W}) = \mathbf{W}$

Shallow Embeddings: Distance-based Non-Euclidean Methods

- Hyperbolic geometry is effective for tree/hierarchical graphs.
- Nickel and Kiela [NK17]: Poincaré model of hyperbolic space.
- Change $d_2(\mathbf{Z}_i, \mathbf{Z}_j)$ to Poincaré distance:

$$d_{\text{Poincare}}(\mathbf{Z}_i, \mathbf{Z}_j) = \operatorname{arcosh}\left(1 + 2 \cdot \frac{\|\mathbf{Z}_i - \mathbf{Z}_j\|_2^2}{\left(1 - \|\mathbf{Z}_i\|_2^2\right)\left(1 - \|\mathbf{Z}_j\|_2^2\right)}\right)$$

- Optimization: minimize connected pairs, maximize disconnected:

$$d_1(\mathbf{W}, \widehat{\mathbf{W}}) = \sum_{i,j} \mathbf{W}_{ij} \log\left(\frac{e^{-\widehat{\mathbf{W}}_{ij}}}{\sum_{k|\mathbf{W}_{ik}=0} e^{-\widehat{\mathbf{W}}_{ik}}}\right)$$

- (Denominator via negative sampling)
- Requires Riemannian optimization [Bon13]
- Other models: Lorentz [NK18], mixed-curvature [Gu+18], hyperbolic skip-gram [CCD17]

Shallow Embeddings: Outer Product-based Matrix Factorization

- Learn low-rank embedding of similarity matrix $s(\mathbf{W})$.
- Common choices: $s(\mathbf{W}) = \mathbf{W}$, Laplacian, Katz, Common Neighbors, Adamic/Adar.
- Decoder: $\widehat{\mathbf{W}} = \mathrm{DEC}(\mathbf{Z}; \mathbf{\Theta}^D) = \mathbf{Z}\mathbf{Z}^\mathsf{T}$.
- Loss: $\mathcal{L}_{G,\mathrm{RECON}} = ||s(\mathbf{W}) - \widehat{\mathbf{W}}||_F^2$.
- Graph Factorization (GF) [Ahm+13]: minimize $\sum_{(v_i,v_j)\in E}(\mathbf{W}_{ij} - \widehat{\mathbf{W}}_{ij})^2$.
- If $\mathbf{A}$ is binary: use sparse loss $||\mathbf{A} \circ (\mathbf{W} - \widehat{\mathbf{W}})||_F^2$.
- GraRep [CLX15]: directed graphs, learns source $\mathbf{Z}^s$ and target $\mathbf{Z}^t$ embeddings.
- k-hop: $\widehat{\mathbf{W}}^{(k)} = \mathbf{Z}^{(k),s}(\mathbf{Z}^{(k),t})^\mathsf{T}$.
- Loss per k: $||\mathbf{D}^{-k}\mathbf{W}^k - \widehat{\mathbf{W}}^{(k)}||_F^2$.
- Final embedding: concatenation across k.
- Not scalable (dense $\mathbf{D}^{-1}\mathbf{W}$), can use implicit matrix factorization [Per+17].

Shallow Embeddings: Outer Product-based Skip-gram Methods (1/2)

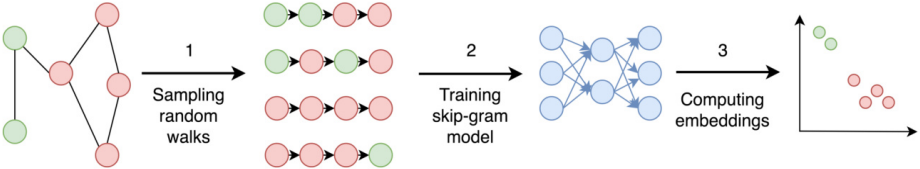
- Inspired by NLP (Word2Vec [Mik+13c; PSM14b]): predict context words for a target word.
- Objective: $\mathcal{L} = -\sum_{-K \leq i \leq K, i \neq 0} \log \mathbb{P}(w_{k-i} \mid w_k)$
- **DeepWalk [PARS14]**:
  - Uses random walks on graphs, mimicking word frequency in language.
  - Decoder: outer product (Eq. 23.13).
  - Generates node sequences via truncated unbiased random walks.
  - Maximizes log-likelihood of predicting context nodes from center node.
  - Uses two representations per node (center and context) [AEHPAR17].
- Transition matrix expectation: $s(\mathbf{W}) = \mathbb{E}_q \left[(\mathbf{D}^{-1}\mathbf{W})^q \right]$, $q \sim \mathrm{Categorical}([1, \ldots, T_{\mathsf{max}}])$

Shallow Embeddings: Outer Product-based Skip-gram Methods (2/2)

- Training minimizes:

$$\mathcal{L}_{G,\text{RECON}}(\mathbf{W}, \widehat{\mathbf{W}}; \Theta) = \log Z(\mathbf{Z}) - \sum_{v_i, v_j} s(\mathbf{W})_{ij} \widehat{\mathbf{W}}_{ij}$$

$$\widehat{\mathbf{W}} = \mathbf{Z}\mathbf{Z}^{\mathsf{T}}, \quad Z(\mathbf{Z}) \text{ via hierarchical softmax or negative sampling}$$



- Figure 23.4: Pipeline for random-walk graph embedding methods.
- Skip-gram as implicit matrix factorization [LG14, Qiu+18] (NetMF framework: DeepWalk, LINE, node2vec).

Shallow Embeddings: Supervised Embeddings (1/2)

- Combines embedding learning with predicting node/graph labels.
- **Label Propagation (LP) [ZG02]**:
  - Semi-supervised node classification.
  - Encoder: shallow lookup table $\mathbf{Z}$.
  - Decoder: identity function $\hat{y}^N = \mathrm{DEC}(\mathbf{Z}; \Theta^C) = \mathbf{Z}$.
  - Smooths label distribution using graph structure (e.g., Laplacian eigenmaps).
  - Assumes neighbor nodes have similar labels; minimizes energy iteratively.
- Reconstruction loss: $\mathcal{L}_{G,\mathrm{RECON}}(\mathbf{W}, \widehat{\mathbf{W}}; \mathbf{\Theta}) = \sum_{i,j} \mathbf{W}_{ij} \|y_i^N - \hat{y}_j^N\|_2^2$

Shallow Embeddings: Supervised Embeddings (2/2)

- **Label Spreading (LS) [Zho+04]**: variant of LP with a different energy function.
- Reconstruction loss:

$$\mathcal{L}_{G,\text{RECON}} = \sum_{i,j} \mathbf{W}_{ij} \left\| \frac{\hat{y}_i^N}{\sqrt{D_i}} - \frac{\hat{y}_j^N}{\sqrt{D_j}} \right\|_2^2$$

- D_i: degree of node v_i.
- Supervised loss (for both LP and LS):

$$\mathcal{L}_{\text{SUP}}^N(y^N, \hat{y}^N; \Theta) = \sum_{i|v_i \in V_L} \|y_i^N - \hat{y}_i^N\|_2^2$$

Graph Neural Networks (GNNs)

- Define convolutions over graph data. [Cha+21]
- (Semi-)supervised neighborhood aggregation methods.
- Encoder: $\mathbf{Z} = \mathrm{ENC}(\mathbf{X}, \mathbf{W}; \Theta^E)$
- Decoders: $\widehat{\mathbf{W}} = \mathrm{DEC}(\mathbf{Z}; \mathbf{\Theta}^D)$ and/or $\widehat{\boldsymbol{y}}^S = \mathrm{DEC}(\mathbf{Z}; \mathbf{\Theta}^S)$

Message Passing GNNs

- Original GNN model [GMS05; Sca+09]: First deep learning for graphs.
- Views supervised graph embedding as information diffusion until equilibrium.
- Recursion with random initial embeddings $\mathbf{Z}^0$:

$$\mathbf{Z}^{t+1} = \mathrm{ENC}(\mathbf{X}, \mathbf{W}, \mathbf{Z}^t; \mathbf{\Theta}^E)$$

- Parameters Θ^E reused.
- After convergence ($t = T$), embeddings used for prediction:

$$\hat{y}^S = \mathrm{DEC}(\mathbf{X}, \mathbf{Z}^T; \mathbf{\Theta}^S)$$

- Parameters learned via backpropagation (Almeda-Pineda algorithm [Alm87; Pin88]).
- Converges if recursion is a contraction mapping (Banach's fixed point theorem).
- Scarselli et al. [Sca+09] use message passing:

$$\mathbf{Z}_i^{t+1} = \sum_{j|(v_i,v_j)\in E} f(\mathbf{X}_i, \mathbf{X}_j, \mathbf{Z}_j^t; \mathbf{\Theta}^E)$$

- $f(\cdot)$ is an MLP constrained to be a contraction mapping.

Message Passing GNNs (Cont.)

- **Gated Graph Sequence Neural Networks (GGSNNs) [Li+15]**:
  - Removes contraction mapping requirement from GNNs.
  - Relaxed recursion: Apply mapping functions (gated recurrent units [Cho+14b]) for fixed steps, parameters shared.
  - Outputs predictions at every step (useful for temporal graphs).
- **Message Passing Neural Networks (MPNNs) [Gil+17]**:
  - Framework for modern multi-layer GNNs with fixed layers.
  - Layer ℓ: Message functions $f^\ell(.)$ receive messages from neighbors, passed to aggregation $h^\ell(.)$:

$$\mathbf{m}_i^{\ell+1} = \sum_{j|(v_i,v_j)\in E} f^\ell(\mathbf{H}_i^\ell, \mathbf{H}_j^\ell)$$

$$\mathbf{H}_i^{\ell+1} = h^\ell(\mathbf{H}_i^\ell, \mathbf{m}_i^{\ell+1})$$

  - $\mathbf{H}^0 = \mathbf{X}$; after ℓ layers, hidden states encode ℓ-hop neighborhoods.
- **GraphNet [Bat+18]**:
  - Extends MPNNs to learn edge, node, and entire graph representations using message passing. Adds expressivity for wider application domains.

Spectral Graph Convolutions

- Define graph convolutions using the spectral domain of the graph Laplacian matrix.
- Two categories:
- **Spectrum-based methods** (e.g., Spectral CNNs [Bru+14]):
  - Explicitly compute eigendecomposition of the Laplacian.
  - Disadvantage: Domain-dependent (cannot generalize to new graphs), computationally expensive.
- **Spectrum-free methods** (e.g., GCN [KW16a]):
  - Motivated by spectral graph theory but no actual spectral decomposition.
  - Utilize approximations of spectral filters.
  - Disadvantage: Require using the whole graph $\mathbf{W}$, do not scale well.
- For more details, see [Bro+17b; Cha+21].

Spatial Graph Convolutions

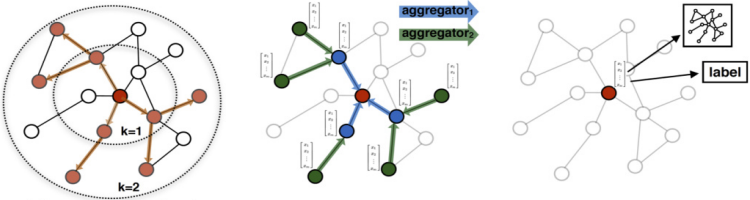
- Overcome limitations of spectral methods (domain dependency, scalability).
- Borrow ideas from standard CNNs: apply convolutions in the spatial domain defined by graph topology.
- Standard CNNs: Filters localized by fixed rectangular patches, pixel ordering allows weight reuse.
- Spatial GCNs: Use neighborhood sampling and attention to create fixed-size graph patches, overcoming irregularity.

Spatial Graph Convolutions: Sampling-based

- **GraphSAGE [HYL17]**: Inductive node embeddings.
  - Instead of averaging all 1-hop neighbors (Laplacian multiplication), SAGE samples fixed-size neighborhoods (q) for each node.
  - Removes dependency on fixed graph structure, allows generalization.
  - At each SAGE layer, nodes aggregate info from sampled neighbors (Figure 23.5).

$$\mathbf{H}_{:,i}^{\ell+1} = \sigma\left(\Theta_1^\ell \mathbf{H}_{:,i}^\ell + \Theta_2^\ell \mathrm{AGG}\left(\left\{\mathbf{H}_{:,j}^\ell \mid v_j \in \mathrm{Sample}(\mathrm{nbr}(v_i), q)\right\}\right)\right)$$

- AGG$(\cdot)$: Permutation invariant operator (e.g., SAGE-mean, SAGE-pool).
- Reduces computational complexity by working with fixed-size neighborhoods.
- Figure 23.5: GraphSAGE model.



1. Sample neighborhood
2. Aggregate feature information
3. Predict graph context and label

Spatial Graph Convolutions: Attention-based

- Attention mechanisms (Sec. 15.4) identify relevant parts of input (e.g., long sequences in language models).
- Applied to GCNs: Learn to focus on important neighbors during message passing via parametric patches learned on node features.
- More flexibility in inductive settings vs. fixed weights (GCNs).
- **Graph Attention Network (GAT) [Vel+18]**:
  - Attention-based GCNs.
  - Attends over neighborhood, selectively picks nodes for downstream task.
  - Similar intuition to SAGE [HYL17], suitable for inductive/transductive problems.
  - Unlike SAGE (fixed-size neighborhoods), GAT attends over all neighbors, assigning different weights.
  - Attention parameters trained via backpropagation, scores row-normalized with softmax.

Spatial Graph Convolutions: Geometric

- **MoNet [Mon+17]**: General framework, works well for features in geometric space (e.g., 3D point clouds, meshes).
- Learns attention patches using parametric functions in a pre-defined spatial domain (e.g., spatial coordinates).
- Applies convolution filters in the resulting graph domain.
- Generalizes methods like Geodesic CNN (GCNN) [Mas+15] and Anisotropic CNN (ACNN) [Bos+16].
- MoNet is more general: any pseudo-coordinates (node features) can induce patches.
- Layer:

$$\mathbf{H}^{\ell+1} = \sigma\left(\sum_{k=1}^{K}(\mathbf{W} \odot g_k(\mathbf{U}^s))\mathbf{H}^\ell \Theta_k^\ell\right)$$

- Gaussian kernel for patches:

$$g_k(\mathbf{U}^s) = \exp\left(-\frac{1}{2}(\mathbf{U}^s - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{U}^s - \boldsymbol{\mu}_k)\right)$$

with $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ learned, $\boldsymbol{\Sigma}_k$ diagonal.

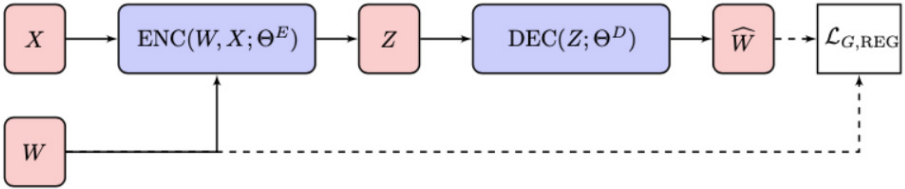
Non-Euclidean Graph Convolutions

- Extends GNNs to learn non-Euclidean embeddings (e.g., hyperbolic for hierarchical graphs [Sec 23.3.3]).
- Challenge: No standard inner product/matrix multiplication in non-Euclidean space.
- **HGCN [Cha+19a]** and **HGNN [LNK19]**:
  - Perform graph convolutions in tangent space (Euclidean approximation at a point).
  - Process: map to tangent space $\rightarrow$ apply convolution $\rightarrow$ map back to hyperbolic.
  - Works well on hierarchical graphs (see Figure 23.6).
- Figure 23.6: Euclidean vs. hyperbolic embeddings of a tree graph.

(a) GCN layers. *(b) HGCN layers.*

Deep Graph Embeddings: Unsupervised

- Using GNNs for unsupervised and semi-supervised graph embeddings.
- Figure 23.7: Structure + features → GNN encoder → embeddings → graph regularization loss.



Unsupervised Deep Embeddings: SDNE

- **SDNE [WCZ16]**: Auto-encoder preserving 1st 2nd order proximity.
- Input: row of adjacency matrix ($s(\mathbf{W}) = \mathbf{W}$), ignores features.
- Output: $\mathbf{Z} = \mathrm{ENC}(\mathbf{W}; \Theta^E)$, then $\widehat{\mathbf{W}} = \mathrm{DEC}(\mathbf{Z}; \Theta^D)$.
- Loss:

$$||(s(\mathbf{W}) - \widehat{\mathbf{W}}) \cdot \mathbb{I}(s(\mathbf{W}) > 0)||_F^2 + \alpha_{\mathrm{SDNE}} \sum_{ij} s(\mathbf{W})_{ij} ||\mathbf{Z}_i - \mathbf{Z}_j||_2^2$$

Unsupervised Deep Embeddings: (V)GAE

- **GAE / VGAE [KW16b]**: Use graph convolutions for encoding.
- $\mathbf{Z} = \mathrm{GCN}(\mathbf{W}, \mathbf{X}; \Theta^E)$, decoder: $\widehat{\mathbf{W}} = \mathbf{Z}\mathbf{Z}^\mathsf{T}$
- Reconstruction loss:

$$\mathcal{L}_{G,\mathrm{RECON}} = -\sum_{i,j} (1 - \mathbf{W}_{ij}) \log(1 - \sigma(\widehat{\mathbf{W}}_{ij})) + \mathbf{W}_{ij} \log \sigma(\widehat{\mathbf{W}}_{ij})$$

- **VGAE**: Uses variational inference with prior $p(\mathbf{Z}) = \mathcal{N}(0, I)$.
- Inference model $q_\Phi(\mathbf{Z}|\mathbf{W}, \mathbf{X})$, minimizes:

$$\mathrm{NELBO} = -\mathbb{E}_q[\log p(\mathbf{W}|\mathbf{Z})] + \mathrm{KL}(q||p)$$

Unsupervised Deep Embeddings: Graphite & Contrastive

- **Graphite [GZE19]**: Iterative decoding + GCN:

$$\widehat{\mathbf{W}}^{(k)} = \frac{\mathbf{Z}^{(k)}\mathbf{Z}^{(k)\mathsf{T}}}{||\mathbf{Z}^{(k)}||_2^2} + \frac{\mathbf{1}\mathbf{1}^{\mathsf{T}}}{N} \quad \mathbf{Z}^{(k+1)} = \mathrm{GCN}(\widehat{\mathbf{W}}^{(k)}, \mathbf{Z}^{(k)})$$

- **DGI [Vel+19]**: Discriminator maximizes mutual info:

$$\min_{\Theta} -\mathbb{E}\sum_i \log \mathcal{D}(\mathbf{Z}_i, \mathcal{R}(\mathbf{Z})) - \sum_j \log(1 - \mathcal{D}(\mathbf{Z}_j^-, \mathcal{R}(\mathbf{Z}^-)))$$

- **GMI [Pen+20]**: Maximizes MI between node and neighborhood embeddings.

Deep Graph Embeddings: Semi-supervised

- **SemiEmb [WRC08]**:
- $\mathbf{Z} = \mathrm{ENC}(\mathbf{X}; \Theta^E)$ via MLP.
- Decoder: $\widehat{\mathbf{W}}_{ij} = ||\mathbf{Z}_i - \mathbf{Z}_j||^2$
- Regularize hidden layers via label propagation loss.
- **Planetoid [YCS16]**: Combines random walks + supervised info.
- $\mathbf{Z} = [\mathbf{Z}^c || \mathbf{Z}^F]$, learn via:

$$\mathcal{L}_{\mathrm{G,RECON}} = -\mathbb{E}_{(i,j,\gamma)} \log \sigma(\gamma \widehat{\mathbf{W}}_{ij}), \quad \widehat{\mathbf{W}}_{ij} = \mathbf{Z}_i \mathbf{Z}_j^{\mathsf{T}}$$

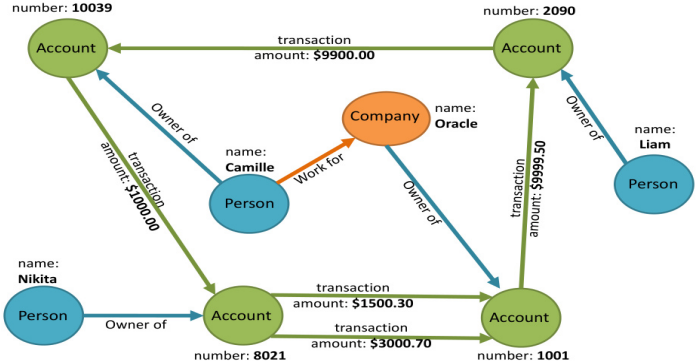
$$\mathcal{L}^N_{\mathrm{SUP}} = -\frac{1}{|V_L|} \sum_{i \in V_L} \sum_k y^N_{ik} \log \widehat{y}^N_{ik}$$

Unsupervised Applications

- **Graph Reconstruction**: Embeddings to reconstruct original graph structure.
- **Link Prediction**: Predict missing or future links.
- Use case: Fraud detection, recommendation, knowledge graphs [Nic+15].

Unsupervised Applications (Cont.)

- Figure 23.8: Graph of financial transactions.



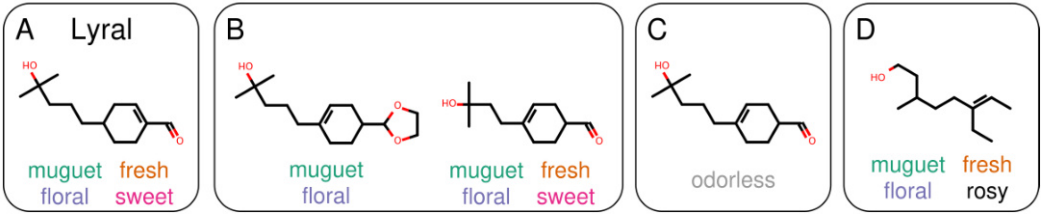
- **Clustering**: Community detection via k-means on embeddings.
- **Visualization**: t-SNE/PCA of embeddings for interpretation.

Supervised Applications: Node Classification

- Predict node labels using learned embeddings.
- **Semi-supervised setting**: Partially labeled graph.
- **Inductive setting**: New nodes/graphs at test time.
- Strong performance from combining structure + features (e.g., GCN, GraphSAGE).

Supervised Applications: Graph Classification

- Predict graph-level labels (e.g., molecule properties).
- Pooling needed to get graph-level features.
- Figure 23.9: Molecules and odor descriptors.



- Can also use unsupervised methods [Tsi+18; ARZP19; TMP20].